

Title: California Housing Price Prediction using Linear Regression

1 Introduction

The objective of this project is to build a Linear Regression model to predict median house values using the California Housing dataset.

The dataset contains housing-related features such as:

- Median income
- Total rooms
- Population
- Latitude & Longitude

The target variable is **median house value**.

2 Exploratory Data Analysis (EDA)

- The dataset contains 20,640 rows and 9 columns.
- No missing values were found.
- Correlation heatmap shows that:
 - **Median Income** has the strongest positive correlation with house value.
 - Some geographical features (latitude/longitude) influence pricing.
- Target variable distribution is slightly right-skewed.

Key Insight:

Median income is the most important feature affecting house prices.

3 Model Building

- Data was split into 80% training and 20% testing.
- Linear Regression model was trained using scikit-learn.
- Predictions were generated on test data.

4 Model Evaluation

The model performance was evaluated using:

- **MAE (Mean Absolute Error)** → Average prediction error
- **RMSE (Root Mean Squared Error)** → Penalizes large errors
- **R² Score** → Measures model accuracy

Interpretation:

- Lower MAE & RMSE indicate better accuracy.
- R² score close to 1 indicates strong model performance.

The model explains a significant portion of variance in housing prices.

5 Improvement Ideas

- Apply **Feature Scaling**
- Try **Ridge / Lasso Regression**
- Use **Polynomial Regression**
- Remove outliers
- Use advanced models (Random Forest, XGBoost)

6 Conclusion

The Linear Regression model successfully predicts house prices with reasonable accuracy.

Further improvements can enhance performance.